UNIT-II

Supervised Learning – I (Regression/Classification)

Regression models: Simple Linear Regression, multiple linear Regression. Cost Function, Gradient Descent, Performance Metrics: Mean Absolute Error(MAE),Mean Squared Error(MSE) R-Squared error, Adjusted R Square.

Classification models: Decision Trees-ID3,CART, Naive Bayes, K-Nearest-Neighbours (KNN), Logistic Regression, Multinomial Logistic Regression Support Vector Machines (SVM) - Nonlinearity and Kernel Methods

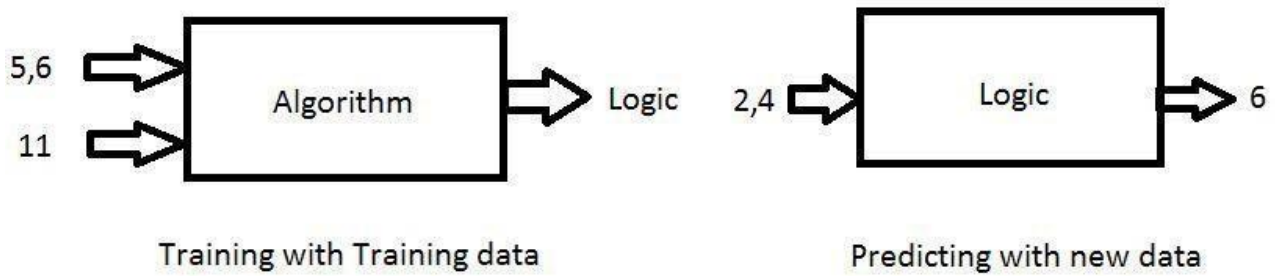
Supervised and unsupervised are mostly used by a lot machine learning engineers and data geeks. Reinforcementlearning is really powerful and complex to apply for problems.

Supervised learning

as we know machine learning takes data as input. lets call this data **Training data**

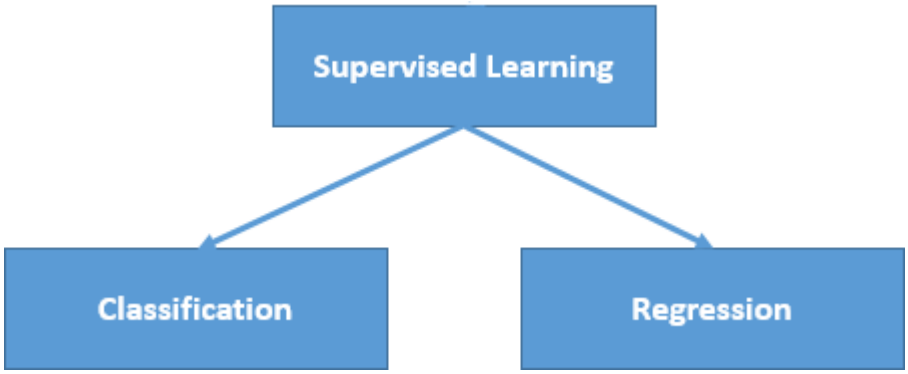
The training data includes both *Inputs* and L*abels(Targets)*

what are Inputs and Labels(Targets)?? for example addition of two numbers a=5,b=6 result =11, *Inputs* are 5,6and *Target* is 11



We first train the model with the lots of training data(inputs&targets)then with new data and the logic we got before we predict the output
(Note :We don't get exact 6 as answer we may get value which is close to 6 based on training data and algorithm)

This process is called *Supervised Learning* which is really fast and accurate.



Types of Supervised learning

Regression: This is a type of problem where we need to predict the *continuous-response* value (ex : above we predictnumber which can vary from -infinity to +infinity)

Some examples are

- what is the price of house in a specific city?
- what is the value of the stock?
- how many total runs can be on board in a cricket game?etc… there are tons of things we can predict if we wish.

Classification: This is a type of problem where we predict the *categorical response* value where the data can beseparated into specific "**classes**" (ex: we predict one of the values in a set of values).

Some examples are :

- this mail is spam or not?
- will it rain today or not?
- is this picture a cat or not?

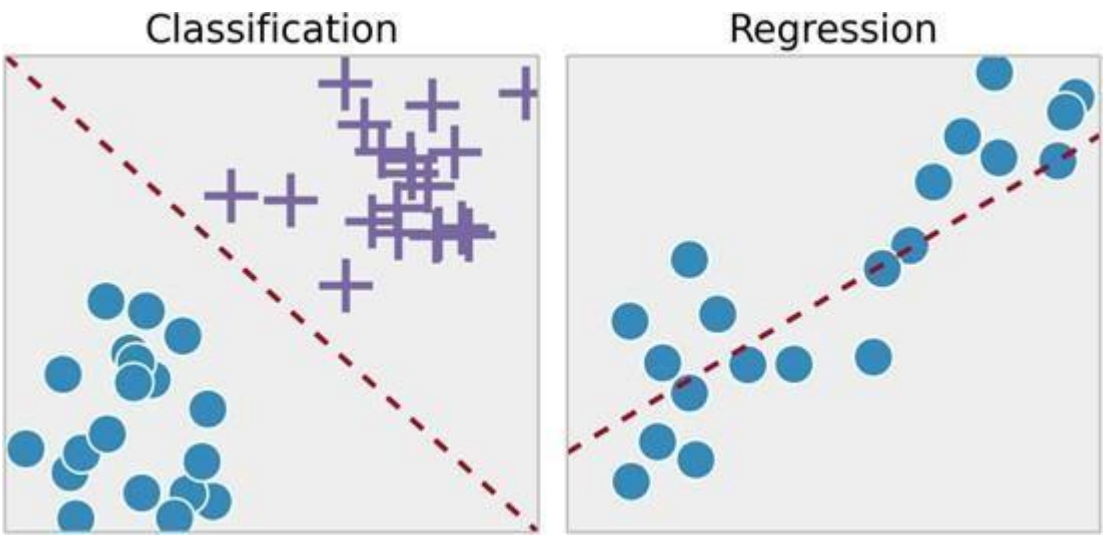
Basically 'Yes/No' type questions called **binary classification.**

Other examples are :

- this mail is spam or important or promotion?
- is this picture a cat or a dog or a tiger?

This type is called **multi-class classification.**

Here is the final picture



Classification separates the data, Regression fits the data

That's all for supervised learning.

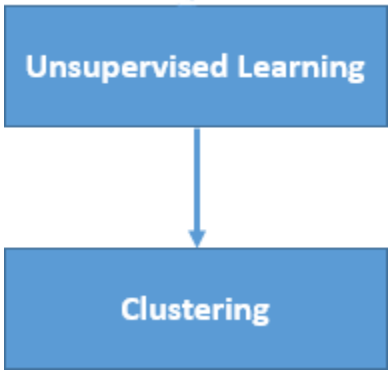
Unsupervised learning

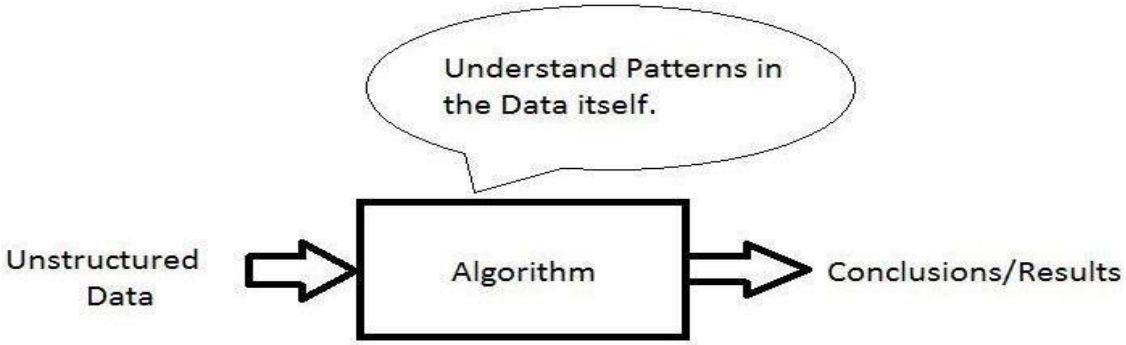
The training data does not include Targets here so we don't tell the system where to go , the system has to understanditself from the data we give.

Here training data is not structured (contains noisy data,unknown data and etc..)

Unsupervised process

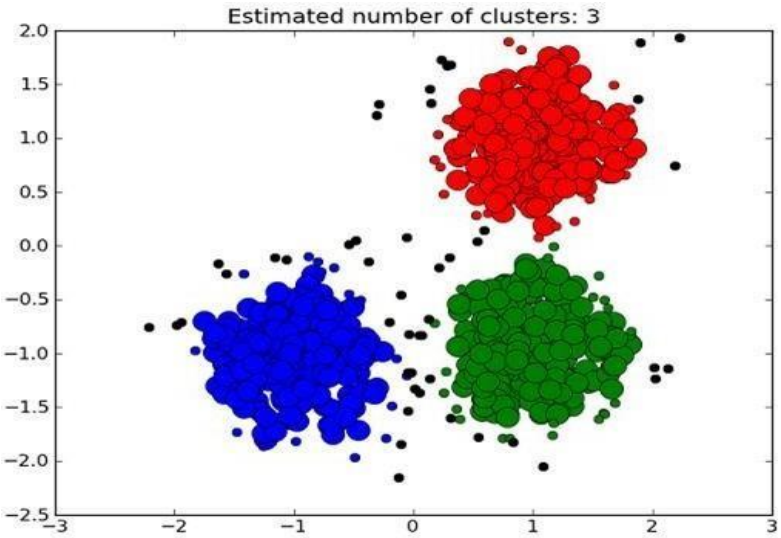
There are also different types for unsupervised learning like Clustering and anomaly detection (clustering is prettyfamous)





rstands

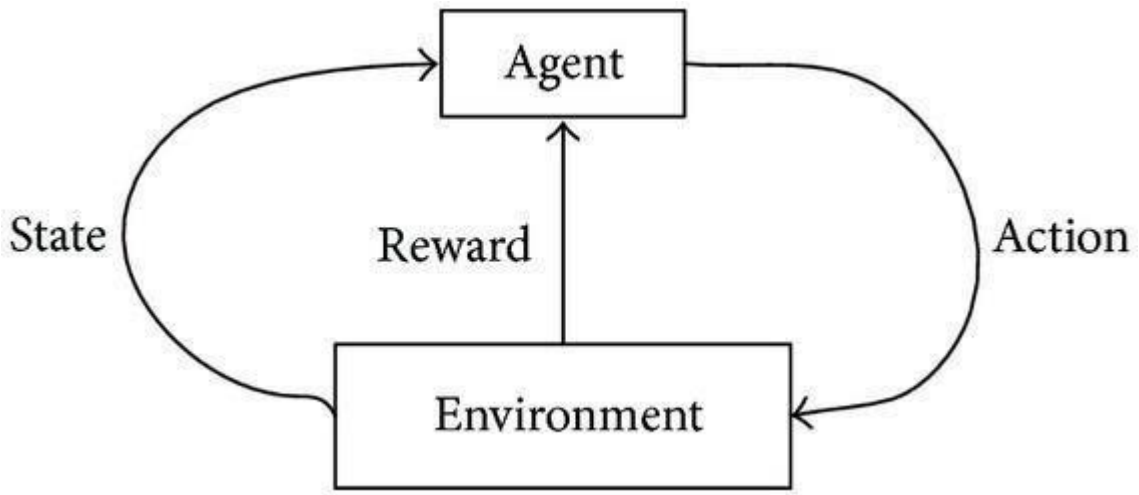
- given a set of tweets ,cluster based on content of tweet
- given a set of images, cluster them into different objects



Clustering with 3 clusters

Unsupervised learning is bit difficult to implement and its not used as widely as supervised.

Reinforcement Learning is a type of *Machine Learning*, and thereby also a branch of *Artificial Intelligence*. It allows machines and software agents to automatically determine the ideal behavior within a specific context, in order to maximize its performance. Simple reward feedback is required for the agent to learn its behavior; this is known as thereinforcement signal.



There are many different algorithms that tackle this issue. As a matter of fact, Reinforcement Learning is defined by a specific type of problem, and all its solutions are classed as Reinforcement Learning algorithms. In the problem, an agent is supposed to decide the best action to select based on his current state. When this step is repeated, the problemis known as a *Markov Decision Process*.

In order to produce intelligent programs (also called agents), reinforcement learning goes through the following steps:

1. Input state is observed by the agent.
2. Decision making function is used to make the agent perform an action.
3. After the action is performed, the agent receives reward or reinforcement from the environment.
4. The state-action pair information about the reward is stored.

List of Common Algorithms

- Q-Learning
- Temporal Difference (TD)

Deep Adversarial Networks

Use cases:

Some applications of the *reinforcement learning algorithms* are computer played board games (Chess, Go), robotic hands, and self-driving cars.